

CA2328: Ensure that JsonSerializerSettings are secure (code analysis)

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Content

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CA2328: Ensure that JsonSerializerSettings are secure

Summarize this article for me

| Property | Value | | | **Rule ID** | CA2328 | | | **Title** | Ensure that JsonSerializerSettings are secure | | | **Category** | [Security](#) | | | **Fix is breaking or non-breaking** | Non-breaking | | | **Enabled by default in .NET 10** | No | | | **Applicable languages** | C# and Visual Basic | |

Cause

This rule fires when both of the following conditions *might be* true for a [Newtonsoft.Json.JsonSerializerSettings](#) instance:

- The [TypeNameHandling](#) property is a value other than `None`.
- The [SerializationBinder](#) property is null.

The [JsonSerializerSettings](#) instance must be used in one of the following ways:

- Initialized as a class field or property.
- Returned by a method.
- Passed to [JsonSerializer.Create](#) or [JsonSerializer.CreateDefault](#).
- Passed to a [JsonConvert](#) method that has a [JsonSerializerSettings](#) parameter.

This rule is similar to [CA2327](#), but in this case, analysis can't definitively determine if the settings are insecure.

By default, this rule analyzes the entire codebase, but this is [configurable]().

Rule description

Insecure deserializers are vulnerable when deserializing untrusted data. An attacker could modify the serialized data to include unexpected types to inject objects with malicious side effects. An attack against an insecure deserializer could, for example, execute commands on the underlying operating system, communicate over the network, or delete files.

This rule finds [Newtonsoft.Json.JsonSerializerSettings](#) instances that might be configured to deserialize types specified from input, but may not be configured to restrict deserialized types with a [Newtonsoft.Json.Serialization.ISerializationBinder](#). If you want to disallow deserialization of types specified from input completely, disable rules [CA2327](#), [CA2328](#), [CA2329](#), and [CA2330](#), and enable rule [CA2326](#) instead.

How to fix violations

- Use [TypeNameHandling](#)'s `None` value, if possible.
- Make the serialized data tamper-proof. After serialization, cryptographically sign the serialized data. Before deserialization, validate the cryptographic signature. Protect the cryptographic key from being disclosed and design for key rotations.
- Restrict deserialized types. Implement a custom [Newtonsoft.Json.Serialization.ISerializationBinder](#). Before deserializing with Json.NET, ensure your custom [ISerializationBinder](#) is specified in the [Newtonsoft.Json.JsonSerializerSettings.SerializationBinder](#) property. In the overridden [Newtonsoft.Json.Serialization.ISerializationBinder.BindToType](#) method, if the type is unexpected, return `null` or throw an exception to stop deserialization.

When to suppress warnings

It's safe to suppress a warning from this rule if:

- You know the input is trusted. Consider that your application's trust boundary and data flows may change over time.
- You've taken one of the precautions in [How to fix violations]().
- You know that the [Newtonsoft.Json.JsonSerializerSettings.SerializationBinder](#) property is always set when [TypeNameHandling](#) property is a value other than `None`.

Suppress a warning

If you just want to suppress a single violation, add preprocessor directives to your source file to disable and then re-enable the rule.

```
#pragma warning disable CA2328
// The code that's violating the rule is on this line.
#pragma warning restore CA2328
```

To disable the rule for a file, folder, or project, set its severity to `none` in the [configuration file](#).

```
[*.{cs,vb}]
dotnet_diagnostic.CA2328.severity = none
```

For more information, see [How to suppress code analysis warnings](#).

Configure code to analyze

Use the following options to configure which parts of your codebase to run this rule on.

- `[Exclude specific symbols]()`
- `[Exclude specific types and their derived types]()`

You can configure these options for just this rule, for all rules they apply to, or for all rules in this category ([Security](#)) that they apply to. For more information, see [Code quality rule configuration options](#).

Exclude specific symbols

You can exclude specific symbols, such as types and methods, from analysis by setting the [excluded_symbol_names](#) option. For example, to specify that the rule should not run on any code within types named `MyType`, add the following key-value pair to an `.editorconfig` file in your project:

```
dotnet_code_quality.CAXXXX.excluded_symbol_names = MyType
```

Note

Replace the `XXXX` part of `CAXXXX` with the ID of the applicable rule.

Allowed symbol name formats in the option value (separated by `|`):

- Symbol name only (includes all symbols with the name, regardless of the containing type or namespace).
- Fully qualified names in the symbol's [documentation ID format](#). Each symbol name requires a symbol-kind prefix, such as `M:` for methods, `T:` for types, and `N:` for namespaces.
- `.ctor` for constructors and `.cctor` for static constructors.

Examples:

Option Value	Summary
<code>dotnet_code_quality.CAXXXX.excluded_symbol_names = MyType</code>	Matches all symbols named <code>MyType</code> .
<code>dotnet_code_quality.CAXXXX.excluded_symbol_names = MyType1 MyType2</code>	Matches all symbols named either <code>MyType1</code> or <code>MyType2</code> .
<code>dotnet_code_quality.CAXXXX.excluded_symbol_names = M:NS.MyType.MyMethod(ParamType)</code>	Matches specific method <code>MyMethod</code> with the specified fully qualified signature.
<code>dotnet_code_quality.CAXXXX.excluded_symbol_names = M:NS1.MyType1.MyMethod1(ParamType) M:NS2.MyType2.MyMethod2(ParamType)</code>	Matches specific methods <code>MyMethod1</code> and <code>MyMethod2</code> with the respective fully qualified signatures.

Exclude specific types and their derived types

You can exclude specific types and their derived types from analysis by setting the [excluded_type_names_with_derived_types](#) option. For example, to specify that the rule should not run on any methods within types named `MyType` and their derived types, add the following key-value pair to an `.editorconfig` file in your project:

```
dotnet_code_quality.CAXXXX.excluded_type_names_with_derived_types = MyType
```

Note

Replace the `XXXX` part of `CAXXXX` with the ID of the applicable rule.

Allowed symbol name formats in the option value (separated by `|`):

- Type name only (includes all types with the name, regardless of the containing type or namespace).
- Fully qualified names in the symbol's [documentation ID format](#), with an optional `T:` prefix.

Examples:

Option value	Summary
<code>dotnet_code_quality.CAXXXX.excluded_type_names_with_derived_types = MyType</code>	Matches all types named <code>MyType</code> and all of their derived types.
<code>dotnet_code_quality.CAXXXX.excluded_type_names_with_derived_types = MyType1 MyType2</code>	Matches all types named either <code>MyType1</code> or <code>MyType2</code> and all of their derived types.
<code>dotnet_code_quality.CAXXXX.excluded_type_names_with_derived_types = M:NS.MyType</code>	Matches specific type <code>MyType</code> with given fully qualified name and all of its derived types.
<code>dotnet_code_quality.CAXXXX.excluded_type_names_with_derived_types = M:NS1.MyType1 M:NS2.MyType2</code>	Matches specific types <code>MyType1</code> and <code>MyType2</code> with the respective fully qualified names, and all of their derived types.

Pseudo-code examples

Violation

```

using System;
using Newtonsoft.Json;
using Newtonsoft.Json.Serialization;

public class BookRecordSerializationBinder : ISerializationBinder
{
    // To maintain backwards compatibility with serialized data before using an
ISerializationBinder.
    private static readonly DefaultSerializationBinder Binder = new
DefaultSerializationBinder();

    public void BindToName(Type serializedType, out string assemblyName, out string
typeName)
    {
        Binder.BindToName(serializedType, out assemblyName, out typeName);
    }

    public Type BindToType(string assemblyName, string typeName)
    {
        // If the type isn't expected, then stop deserialization.
        if (typeName != "BookRecord" && typeName != "AisleLocation" && typeName !=
"WarehouseLocation")
        {
            return null;
        }

        return Binder.BindToType(assemblyName, typeName);
    }
}

public class BookRecord
{
    public string Title { get; set; }
    public object Location { get; set; }
}

public abstract class Location
{
    public string StoreId { get; set; }
}

public class AisleLocation : Location
{
    public char Aisle { get; set; }
}

```

```
    public byte Shelf { get; set; }
}

public class WarehouseLocation : Location
{
    public string Bay { get; set; }
    public byte Shelf { get; set; }
}

public static class Binders
{
    public static ISerializationBinder BookRecord = new BookRecordSerializationBinder();
}

public class ExampleClass
{
    public BookRecord DeserializeBookRecord(string s)
    {
        JsonSerializerSettings settings = new JsonSerializerSettings();
        settings.TypeNameHandling = TypeNameHandling.Auto;
        settings.SerializationBinder = Binders.BookRecord;
        return JsonConvert.DeserializeObject<BookRecord>(s, settings);    // CA2328 --
        settings might be null
    }
}
```

```
Imports System
Imports Newtonsoft.Json
Imports Newtonsoft.Json.Serialization
```

```
Public Class BookRecordSerializationBinder
```

```
    Implements ISerializationBinder
```

```
    ' To maintain backwards compatibility with serialized data before using an
    ISerializationBinder.
```

```
    Private Shared ReadOnly Property Binder As New DefaultSerializationBinder()
```

```
    Public Sub BindToName(serializedType As Type, ByRef assemblyName As String, ByRef
typeName As String) Implements ISerializationBinder.BindToName
```

```
        Binder.BindToName(serializedType, assemblyName, typeName)
```

```
    End Sub
```

```
    Public Function BindToType(assemblyName As String, typeName As String) As Type
    Implements ISerializationBinder.BindToType
```

```
        ' If the type isn't expected, then stop deserialization.
```

```
        If typeName <> "BookRecord" AndAlso typeName <> "AisleLocation" AndAlso typeName
<> "WarehouseLocation" Then
```

```
            Return Nothing
```

```
        End If
```

```
        Return Binder.BindToType(assemblyName, typeName)
```

```
    End Function
```

```
End Class
```

```
Public Class BookRecord
```

```
    Public Property Title As String
```

```
    Public Property Location As Location
```

```
End Class
```

```
Public MustInherit Class Location
```

```
    Public Property StoreId As String
```

```
End Class
```

```
Public Class AisleLocation
```

```
    Inherits Location
```

```
    Public Property Aisle As Char
```

```
    Public Property Shelf As Byte
```

```
End Class
```

```

Public Class WarehouseLocation
    Inherits Location

    Public Property Bay As String
    Public Property Shelf As Byte
End Class

Public Class Binders
    Public Shared Property BookRecord As ISerializationBinder = New
    BookRecordSerializationBinder()
End Class

Public Class ExampleClass
    Public Function DeserializeBookRecord(s As String) As BookRecord
        Dim settings As JsonSerializerSettings = New JsonSerializerSettings()
        settings.TypeNameHandling = TypeNameHandling.Auto
        settings.SerializationBinder = Binders.BookRecord
        Return JsonConvert.DeserializeObject(Of BookRecord)(s, settings) ' CA2328 --
        settings might be Nothing
    End Function
End Class

```

Solution

```

using System;
using Newtonsoft.Json;
using Newtonsoft.Json.Serialization;

public class BookRecordSerializationBinder : ISerializationBinder
{
    // To maintain backwards compatibility with serialized data before using an
ISerializationBinder.
    private static readonly DefaultSerializationBinder Binder = new
DefaultSerializationBinder();

    public void BindToName(Type serializedType, out string assemblyName, out string
typeName)
    {
        Binder.BindToName(serializedType, out assemblyName, out typeName);
    }

    public Type BindToType(string assemblyName, string typeName)
    {
        // If the type isn't expected, then stop deserialization.
        if (typeName != "BookRecord" && typeName != "AisleLocation" && typeName !=
"WarehouseLocation")
        {
            return null;
        }

        return Binder.BindToType(assemblyName, typeName);
    }
}

public class BookRecord
{
    public string Title { get; set; }
    public object Location { get; set; }
}

public abstract class Location
{
    public string StoreId { get; set; }
}

public class AisleLocation : Location
{
    public char Aisle { get; set; }
}

```

```

    public byte Shelf { get; set; }
}

public class WarehouseLocation : Location
{
    public string Bay { get; set; }
    public byte Shelf { get; set; }
}

public static class Binders
{
    public static ISerializationBinder BookRecord = new BookRecordSerializationBinder();
}

public class ExampleClass
{
    public BookRecord DeserializeBookRecord(string s)
    {
        JsonSerializerSettings settings = new JsonSerializerSettings();
        settings.TypeNameHandling = TypeNameHandling.Auto;

        // Ensure that SerializationBinder is non-null before deserializing
        settings.SerializationBinder = Binders.BookRecord ?? throw new
Exception("Expected non-null");

        return JsonConvert.DeserializeObject<BookRecord>(s, settings);
    }
}

```

```
Imports System
Imports Newtonsoft.Json
Imports Newtonsoft.Json.Serialization
```

```
Public Class BookRecordSerializationBinder
```

```
    Implements ISerializationBinder
```

```
    ' To maintain backwards compatibility with serialized data before using an
    ISerializationBinder.
```

```
    Private Shared ReadOnly Property Binder As New DefaultSerializationBinder()
```

```
    Public Sub BindToName(serializedType As Type, ByRef assemblyName As String, ByRef
    typeName As String) Implements ISerializationBinder.BindToName
```

```
        Binder.BindToName(serializedType, assemblyName, typeName)
```

```
    End Sub
```

```
    Public Function BindToType(assemblyName As String, typeName As String) As Type
    Implements ISerializationBinder.BindToType
```

```
        ' If the type isn't expected, then stop deserialization.
```

```
        If typeName <> "BookRecord" AndAlso typeName <> "AisleLocation" AndAlso typeName
    <> "WarehouseLocation" Then
```

```
            Return Nothing
```

```
        End If
```

```
        Return Binder.BindToType(assemblyName, typeName)
```

```
    End Function
```

```
End Class
```

```
Public Class BookRecord
```

```
    Public Property Title As String
```

```
    Public Property Location As Location
```

```
End Class
```

```
Public MustInherit Class Location
```

```
    Public Property StoreId As String
```

```
End Class
```

```
Public Class AisleLocation
```

```
    Inherits Location
```

```
    Public Property Aisle As Char
```

```
    Public Property Shelf As Byte
```

```
End Class
```

```

Public Class WarehouseLocation
    Inherits Location

    Public Property Bay As String
    Public Property Shelf As Byte
End Class

Public Class Binders
    Public Shared Property BookRecord As ISerializationBinder = New
    BookRecordSerializationBinder()
End Class

Public Class ExampleClass
    Public Function DeserializeBookRecord(s As String) As BookRecord
        Dim settings As JsonSerializerSettings = New JsonSerializerSettings()
        settings.TypeNameHandling = TypeNameHandling.Auto

        ' Ensure that SerializationBinder is non-null before deserializing
        settings.SerializationBinder = If(Binders.BookRecord, New Exception("Expected
non-null"))

        Return JsonConvert.DeserializeObject(Of BookRecord)(s, settings)
    End Function
End Class

```

CA2326: Do not use TypeNameHandling values other than None

CA2327: Do not use insecure JsonSerializerSettings

CA2329: Do not deserialize with JsonSerializer using an insecure configuration

CA2330: Ensure that JsonSerializer has a secure configuration when deserializing